

Lucio Ruben Villena

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EDUCATION

University of Central Florida – Orlando, FL

M.S. Computer Engineering

University of Central Florida – Orlando, FL

May 2026

B.S. Computer Engineering GPA: 3.62

Relevant Coursework:

- ❖ Completed: Data Structures and Algorithms I and II, Algorithms for Machine Learning, Processes for Object Oriented Development, Object Oriented Programming, Computer Organization, Embedded Systems

TECHNICAL SKILLS

- ❖ **Languages:** Python, C/C++, Java, JavaScript, SQL
- ❖ **Frameworks:** React, Node.js, Express, Tailwind CSS
- ❖ **ML/Data:** NumPy, Pandas, scikit-learn, PyTorch, HuggingFace Transformers, PEFT/QLoRA, TRL, BitsAndBytes, sacrebleu
- ❖ **Systems:** Linux, Git, Nginx, PM2, Google Colab, Docker, AWS

PROJECTS

S.T.A.R. —Spotlight Tracking Automated Recognition— Raspberry Pi CM5, ESP32-S3, YOLO26n, Python, C/C++, React Native [Github](#)

- ❖ **1st Place — UCF CREOL (College of Optics and Photonics) Senior Design Showcase** - Built a spotlight that utilizes a CMOS camera to automatically track a performer on stage using Computer Vision
- ❖ Engineered a real-time computer vision pipeline achieving 15–30 FPS (33–66 ms/frame) using YOLO26n on Raspberry Pi CM5 to detect and localize human subjects across a 7.5 m range and >45° FOV; end-to-end system latency of 392 ms (below 500 ms requirement) with 77% avg detection confidence across 10 trials.
- ❖ Implemented low-latency UART communication (up to 1 Mbps) streaming normalized coordinates to an ESP32-S3 controlling a dual-axis servo gimbal via PCA9685 PWM; gimbal range: 180° pan / 120° tilt at ≥180°/s pan, ≥120°/s tilt.
- ❖ Developed a React Native mobile app for manual override, calibration, and real-time telemetry via WebSocket for low-latency bi-directional communication.

Recipe Generation via QLoRA Fine-Tuning — Gemma 2B, PyTorch, HuggingFace, PEFT/QLoRA, Python [Github](#)

- ❖ Fine-tuned Google's Gemma 2B-it on a subset of Food.com recipes using QLoRA 4-bit NF4 quantization (BitsAndBytes) across 3 LoRA ranks (r=4, 16, 64) to generate structured recipes from ingredient lists; trained on the UCF Newton GPU Cluster via SLURM job scheduling
- ❖ Conducted a full ablation study — zero-shot Gemma-2B baseline, TF-IDF K-Nearest Neighbors retrieval baseline, and 3 QLoRA fine-tuning runs — evaluated on held-out examples using corpus ROUGE-1, ROUGE-2, and ROUGE-L scores, with all results logged to a structured CSV for comparison.
- ❖ Built a modular training pipeline with shared preprocessing, prompt formatting, and 90/5/5 train/val/test splits; trained with Paged AdamW (8-bit), lr=2e-4, and a cosine scheduler, with the full experimental pipeline orchestrated end-to-end by a single SLURM submission script.
- ❖ Designed an interactive terminal inference CLI supporting configurable adapter loading by rank for real-time recipe generation; structured outputs and sample generations saved per run to ablation_outpts/ for reproducible analysis.

travelinggenie.com — AI Trip Planner — React, Node, Express, MongoDB, Gemini API, Linux [Github](#)

- ❖ Developed a full-stack AI-powered itinerary generation platform using React frontend and Node.js/Express backend; integrated Gemini LLM API to generate personalized travel plans from structured user inputs.
- ❖ Designed RESTful APIs for authentication and itinerary CRUD operations, backed by MongoDB with indexed queries; deployed to a Linux VM with Nginx reverse proxy and PM2 process management.
- ❖ Collaborated in a 7-developer Agile team using Git branching, pull requests, and code reviews to maintain code quality.

LEADERSHIP

Pi Delta Psi Fraternity Inc. – Orlando, FL

❖ **President**

May 2025 – April 2026

- Led a **25-member** organization as chief executive, driving strategic planning, cross-functional coordination, and university compliance to sustain chapter operations and growth.
- Managed stakeholder communication across national leadership and university administration; made time-sensitive decisions balancing risk, budget constraints, and team needs under ambiguity.
- Grew chapter membership by **83%** by redesigning the recruitment and onboarding pipeline, streamlining intake of **15** new members and earning the national Outstanding Recruitment award.
- Planned and executed **10+** fundraising and community events generating **\$7,780** in revenue; chapter awarded **Fall 2025 Chapter of Excellence** by the national organization.

ADDITIONAL

- ❖ **Honors:** Dean's/President's List (multiple terms).